ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO TECHNICAL PROGRAMMING

PAPER 1 · PRACTICE E

MARKS: 120 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET first paper (120 marks). Section A (31): multiple choice, true or false, matching C++ terms and naming the parts of a for loop · Section B (89): computer and programming concepts, trace tables and naming rules, computational thinking for a recycling robot, an mBlock temperature program to study (with C++ and a flowchart) and a C++ stock-search program with arrays.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Program code in this paper is written in C++, except for the mBlock program in QUESTION 7. Code is not marked strictly on syntax.
6. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Which device is used to type text into a computer?

A Monitor
B Keyboard
C Speaker
D Printer

1.2 What does #include <iostream> do in a C++ program?

A It starts the main function.
B It ends the program.
C It declares a variable.
D It adds the library for input and output.

1.3 What is displayed by: int x = 5; x += 3; cout << x;

A 8
B 5
C 3
D 53

1.4 Which C++ data type stores the text "Hello"?

A int
B char
C string
D bool

1.5 mBlock is based on …

A Scratch.
B Excel.
C Python only.
D HTML.

1.6 Which Git command creates a new, empty repository in the current folder?

A git clone
B git init
C git push
D git log

1.7 Which symbols are the logical OR operator in C++?

A &&
B !
C ||
D ==

1.8 The value of 17 % 5 in C++ is …

A 3.
B 3.4.
C 12.
D 2.

1.9 The first step of Polya's problem-solving method is to …

A understand the problem.
B carry out the plan.
C look back.
D devise a plan.

1.10 A trace table is used to …

A draw a circuit.
B follow the values of variables step by step.
C store files.
D compile code.

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.5) in the ANSWER BOOK.

2.1 C++ is a compiled language.

2.2 In C++, a variable must be declared before it is used.

2.3 The statement x = x + 1 makes the value of x smaller.

2.4 git pull downloads the latest changes from a remote repository.

2.5 In mBlock, the "when green flag clicked" block must be placed at the end of a script.

(5)

[5]

QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.10) in the ANSWER BOOK.

	COLUMN A		COLUMN B
3.1	Data type for decimal numbers with double precision	A	switch
3.2	Reads a whole line of text, including spaces	B	.length()
3.3	Gives the number of characters in a string	C	double
3.4	Stores several values of the same type, found by an index	D	return
3.5	Adds 1 to a variable	E	static_cast
3.6	Greater than or equal to	F	array
3.7	Ends a function and sends back a value	G	getline()
3.8	A named block of code that can be called	H	function
3.9	Chooses one of several cases using a value	I	>=
3.10	Changes a value into another data type	J	++

(10)

[10]

QUESTION 4

Study the C++ loop below. Name the part of the loop on each line marked A–F (4.1–4.6).

```
A   for (int i = 0;
B         i < 5;
C         i++)
D   {
E       cout << i << endl;
F   }
```

4.1 A

4.2 B

4.3 C

4.4 D

4.5 E

4.6 F

(6)

[6]

TOTAL SECTION A: [31]

SECTION B

QUESTION 5

5.1 Answer the questions below. Write only the answer next to the question number (5.1.1–5.1.5).

5.1.1 Define the term integrated development environment (IDE). (2)

5.1.2 Define the term data type, and give TWO examples of C++ data types. (4)

5.1.3 Which type of memory stores the start-up instructions (firmware) of a computer? (2)

5.1.4 Which part of the CPU carries out calculations and comparisons? (2)

5.1.5 Define the term compiler. (2)

5.2 Define the term pattern recognition in computational thinking. (2)

5.3 Explain the difference between a while loop and a for loop. (4)

5.4 Explain what a trace table is and why programmers use it. (4)

5.5 Give FOUR rules for naming variables in C++. (4)

[26]

QUESTION 6

Answer the following question as fully as possible.

> A municipality wants a robot that sorts waste on a conveyor belt into plastic, glass, paper and metal. The robot has a camera, a metal sensor and a robot arm that drops each item into the correct bin. You are part of the team that must plan the program.

Explain how the FOUR cornerstones of computational thinking (decomposition, pattern recognition, abstraction and algorithm design) can be used to plan the program for this recycling robot. Refer to the robot in each part of your answer. (11)

[11]

QUESTION 7

Study the mBlock program below and answer the questions.

```
when green flag clicked
set [total] to 0
set [count] to 0
set [highest] to 0
repeat 5
    ask [Enter today's temperature] and wait
    set [temp] to (answer)
```

```
    change [total] by (temp)
    change [count] by 1
    if < (temp) > (highest) > then
        set [highest] to (temp)
end repeat
set [average] to ((total) / (count))
if < (average) > 25 > then
    say [Hot week] for 2 seconds
else
    say [Cool week] for 2 seconds
say (join [Highest: ] (highest)) for 2 seconds
```

7.1 Name FOUR variables used in the program and say what each one stores. (4)

7.2 What does the block "change total by temp" do? (2)

7.3 Why is highest set to 0 at the start, and when could this cause a wrong answer? (3)

7.4 The user enters 20, 28, 31, 22 and 24. Calculate the average and write down both messages that the sprite says. (4)

7.5 Write a C++ program that does the same as the mBlock program, using a for loop. (8)

7.6 Explain why a for loop is a good choice for this problem. (2)

7.7 Draw a flowchart for the mBlock program above. (9)

[32]

QUESTION 8

A robotics workshop keeps the names of five parts in one array and the number of each part in stock in a second array. A technician types the name of a part to find out how many are left.

Study the program below and answer the questions that follow.

```
 1  #include <iostream>
 2  #include <string>
 3  using namespace std;
 4
 5  int main() {
 6      string parts[5] = {"motor", "sensor", "wheel", "battery", "servo"};
 7      int stock[5] = {12, 4, 20, 0, 7};
 8      string wanted;
 9      bool found = false;
10
11      cout << "Which part? ";
12      cin >> wanted;
13      for (int i = 0; i < 5; i++) {
14          if (parts[i] == wanted) {
15              found = true;
16              if (stock[i] == 0) {
17                  cout << wanted << " is out of stock" << endl;
18              } else if (stock[i] < 5) {
19                  cout << "Only " << stock[i] << " left - reorder soon" << endl;
20              } else {
21                  cout << stock[i] << " in stock" << endl;
22              }
23          }
```

```
24      }
25      if (!found) {
26          cout << "Unknown part" << endl;
27      }
28      return 0;
29  }
```

8.1 How many elements does each array hold, and how are their values given? (2)

8.2 What is the index of "wheel" in the parts array? (2)

8.3 What is displayed if the technician enters battery? (2)

8.4 What is displayed if the technician enters sensor? (2)

8.5 What is displayed if the technician enters servo? (2)

8.6 What is displayed if the technician enters gripper? (2)

8.7 What is the purpose of the variable found? (2)

8.8 What does the ! operator do in line 25? (2)

8.9 Why is the string library included in line 2? (2)

8.10 Suggest how the loop could be made to stop as soon as the part has been found. (2)

[20]

TOTAL SECTION B: [89]

TOTAL: 120

END OF QUESTION PAPER